CNEC Critical Thinking | 26

Port Automation and Risks:

Problem, Case, Policy recommendation

AS0092

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1. Introduction & Research Question

Handled 10 million twenty-foot cargoes by February 2025, the world's largest fully automated container terminal, Tuas Port in Singapore, attracts people's attention to the maritime industry and its automation. (PSA Singapore, 2025) Regarded as the backbone of globalization, the maritime industry transports more than 80% of global trade. (UNCTAD, 2025)

Tuas Port is one of 63 semi- or fully automated ports worldwide (Knatz et al., 2022), and the trend toward automation continues to grow. The trend, however, creates structural shifts that pose challenges, namely market concentration, raising concerns for policymakers. Using microeconomic analysis through deductive reasoning, this essay examines how automation drives market concentration, analyzes the resulting structural shift and welfare impact, and recommends relevant policy.

2. Key concepts & Model and Analytical Framework

The essay's key economic concepts are market power, which is central to the analysis of automated ports. Market power, or pricing power, refers to a firm's ability to raise and sustain prices above the allocative-efficiency level, where price exceeds marginal cost (P > MC), typically arising in highly concentrated markets with substantial barriers to entry. In this case, automation confers market power on PSA Singapore, enabling it to charge high port charges, thereby leading to several issues.

The essay adopts economies of scale and network externalities as its primary models. Economies of scale explain how high fixed costs of automation can be spread over large volumes of transshipment, lowering average costs for dominant ports. Network externalities illustrate how the value of a transshipment hub increases as more shipping lines and routes concentrate in the same port, resulting in a positive feedback loop.

Besides, this essay utilizes the Structure–Conduct–Performance (SCP) framework as its primary analytical lens.

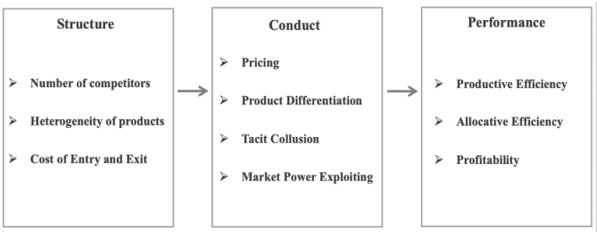


Figure 1: Diagram for SCP Paradigm

The SCP framework is applied by first analyzing how port automation alters the market structure of Southeast Asia's container transshipment market, leading to high market concentration, and then examining how this structural change affects firm conduct, focusing on pricing behavior under conditions of market power and information asymmetry. Finally, the framework is used to assess market performance, with particular attention to allocative efficiency and social welfare.

3. Case Study

3.1 Theoretical Mechanism

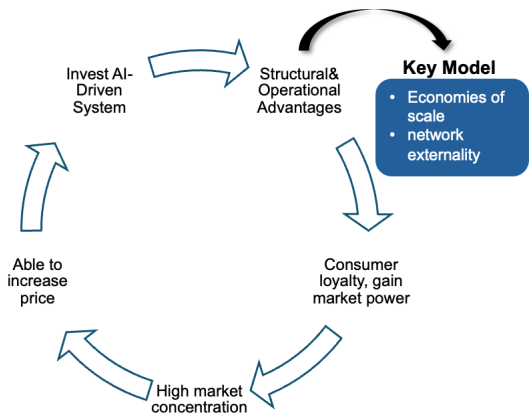


Figure 2: Diagram of Mechanism.

Figure 2 visualizes the mechanism overview of the Singapore Port case study.

Specifically, by investing in an automation system, the port can gain both a structural advantage, including economies of scale and network externalities, and operational advantages such as time& cost advantage, reliability advantage, and sustainability advantage, and thus attract more shipping companies and create Cargo throughput concentration and Hub- Lock in effect. This confers market power on the Singapore port, allowing it to charge high prices, thereby generating large welfare losses and allocative inefficiency. With a higher price, the port can also invest more in AI transformation, creating a positive feedback loop.

3.2 Case Analysis: PSA Singapore

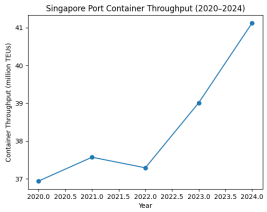
3.2.1 Technology Background

The Singapore Tuas port is equipped with a fully automated operational system driven by event-driven architecture (EDA). Equipped with automated storage and retrieval systems, the port can handle the transport of pallets on a 24/7 basis. In addition, the data exchange platform integrates IoT sensors and video cameras to monitor traffic conditions and improve asset utilization. Sorting systems and active inventory tracking are also automated to power rapid order fulfillment and management. (Williams, 2025) Owning the largest fleet of 400 automated guided vehicles, the port automatically transports cargoes without rest. (Hand, 2026)

3.2.2 Operational Advantages of Tuas Port

The primary drivers of the structural change are operational advantages, which include time and cost, stability, and sustainability, and structural advantages involving economies of scale and network externality.

First, the EDA platforms-driven automated systems in Singapore ports empower ports to optimize their resource allocation, make informed decisions ahead of time, and effectively streamline workflows– ultimately improving handling efficiency and scheduling precision and reducing manual delays. This limits the time and cost to process each throughput(Tsagkaris & Moschovou, 2025)



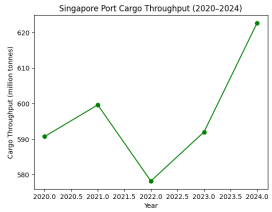


Figure 3: Diagram of Singapore Container Throughput

Figure 4: Diagram of Singapore Cargo Throughput

Figures 3 and 4 illustrate the trends in container throughput and total cargo throughput at the Port of Singapore from 2020 to 2024. Container throughput increased steadily from 36.94 million TEUs in 2020 to 41.12 million TEUs in 2024, and Cargo throughput increased from 590.74 million in 2020 to 620.67 million in 2024 (*ABOUT MPA*, n.d.), (*MPA Integrated Annual Report 2022*, 2022), indicating a sustained improvement in total handling capacity. Rising throughput volume in a single year suggests higher operational efficiency and a shorter average processing time per TEUs.

Moreover, although the absolute value of operating costs is increasing, this is attributable to investments in port expansion and substantial capital expenditures for automation. By dividing the operating costs by container throughput, the operating cost per TEU can be calculated:

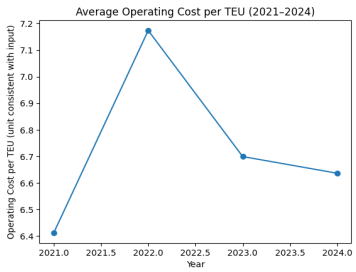


Figure 5: Diagram of average operating cost per TEU

As shown in Figure 5, unit operating cost of ports in Singapore peaked in 2022 and subsequently declined in 2023–2024 as container throughput expanded(*ABOUT MPA*, n.d.), (*MPA*

Integrated Annual Report 2022, 2022). This suggests that efficiency gains primarily from automation lead to lower cost per TEU. Overall, the cost and time advantages attract more shipping traffic and lead to throughput concentration, where cargo increasingly flows through these efficient hubs.

Meanwhile, AI systems reduce variability and unexpected downtime through predictive analytics, mathematical modeling, and real-time adaptability. This capability ensures stable performance, reduces risks and uncertainty for shipping companies, which value schedule reliability as a key component of service quality(Kottara, 2025). As a result, shipping lines are more likely to integrate such ports into their long-term network planning and hub selection, increasing switching costs and creating stable demand for the port. The importance of operational stability became particularly evident during the COVID 19 period from 2020 to 2022. During that period, when many global ports like the Port of Los Angeles and Port of Long Beach experienced severe congestion and operational disruptions due to labor shortages and health restrictions(UNCTAD, 2021), the Port of Singapore maintained operations 24/7 and avoided prolonged shutdowns because of its low dependence on on-site labor and automation's ability to reallocate resources and adjust coordination(*PSA SINGAPORE*, n.d.). Following severe disruptions, schedule reliability is a critical determinant of hub selection in global shipping networks. Thus, Singapore ports have established a reputation for safety and stability, making them increasingly preferred by shipping companies.

Additionally, a report mentions that Tuas Port will deploy electrified equipment and vehicles, such as Automated Guided Vehicles (AGVs), to reduce carbon emissions("Environmental Benefits of Port Automation," 2025). As global trade increasingly intersects with environmental regulation, ports with strong sustainability performance can lower compliance costs and attract shipping lines that value green logistics. Sustainability thus becomes a competitive advantage that can attract more shipping lines, experience compliance costs, and build corporate social responsibility

3.2.3 Structural Advantages of Tuas Port

Considering structural advantages, Singapore's port's leading role stems from its outstanding location and infrastructure. Its location in the Strait of Malacca provides a natural advantage, connecting it to more than 600 ports and 200 shipping lines. (MPA., 2026) It also has the best infrastructure worldwide with high efficiency. In recent years, Singapore Port's dominance has declined due to rapid growth at smaller ports, with the container port market in Southeast Asia shifting from highly concentrated to moderately concentrated. However, with the trend toward automation, market concentration is likely to increase again.

The automation of ports will further strengthen Singapore Port's dominant role in Southeast Asia, owing to the network externalities of AI technologies. Automated ports have lower average waiting time, higher loading and unloading efficiency, more stable services and broader coverage than ordinary ports. For instance, inefficiencies due to complex paperwork and low customs clearance efficiency can be mitigated by blockchain technology. (Li et al., 2023) These traits can attract more shippers and create a positive feedback loop, continuously amplifying its competitive edge. Maritime transshipment will be more and more concentrated in these ports. Maritime transshipment will be increasingly concentrated in leading automated ports and the gap with surrounding rivals will be widened.

Another reason Singapore Port may become more dominant is economies of scale. The process of automating a port requires substantial investment and R&D costs, which are common across ports; however, top ports

such as Singapore Port handle massive throughput, spreading these costs across more cargo units. Therefore, even though smaller ports are automated, their port dues must be higher than Singapore ports, making them less competitive.

3.2.4 High Concentration Market

Overall, Singapore's early adoption of large-scale port automation generated a first-mover advantage in the Southeast Asian transshipment market. By achieving efficiency, reliability, and scale benefits earlier than regional competitors, Singapore attracted a significant share of shipping lines and established itself as the primary transshipment hub (Notteboom & Yim Yap, n.d.).

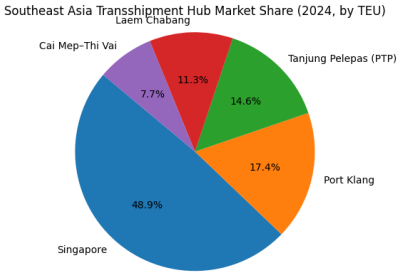


Figure 5: Market share Pie Chart

As shown in Figure 5, Singapore captures nearly half of the Southeast Asian transshipment hub market, indicating a highly concentrated market structure(World Bank Group, 2024). While other ports, such as Port Klang, Tanjung Pelepas, Laem Chabang, and the Cai Mep–Thi Vai cluster, have begun adopting automation and AI systems in recent years， and their throughput shares have gradually increased.

3.2.5 Critical problem behind the highly concentrated market

Such high market concentration offers significant market power to Singapore ports, enabling greater pricing power over port charges and strengthening hub lock-in effects through higher switching costs for shipping lines.

The price of using Singapore Ports is higher than that of nearby ports in Malaysia, Thailand, and other countries, and it continues to rise. For instance, on 1 July 2022 and 2023, the port raised stevedoring recovery charges by about 7.5% each time. (X-Press Feeders, 2022) Higher transaction costs erode exporters' profit margins and surpluses and may lead to a decline in trade volume. On the social allocative efficiency front, rising prices lead to underconsumption of port services and greater total welfare loss. Meanwhile, high market concentration may result in idle resources at smaller ports, thereby further amplifying allocative inefficiency.

At the industry level, a high concentration ratio may inhibit further innovation. In the short run, automation of PSA port yields short-term gains from the boosted trading efficiency, but a possible monopoly reduces incentives for ongoing R&D if the port faces low competition. Unlike ports in competitive markets that must rely on innovation to sustain advantages, PSA can maintain its dominant position through existing scale and automation barriers, thereby slowing the overall rate of technological innovation in the regional port industry. The ports may incur X-inefficiency, in which actual operating costs exceed the average cost, if there is a lack of competition. It may take a long period, with government intervention in the areas of monopoly and innovation, for the industry to further develop.

At the macro level, a higher concentration ratio in the port industry may exacerbate systemic vulnerabilities in shipping and trade. If a large proportion of cargo transshipment in a region concentrates in a few ports, the negative impact of unexpected risks will be even worse. Heavily reliant on digital systems and AI scheduling, the port faces vulnerabilities from cyberattacks, power outages or system failures—any disruption would halt cargo transportation and vessel operations instantly.

For instance, before the competition of AI systems, Singapore's automated Tuas Port had processed over 10 million TEUs, while after

completion, it handled 41.1 million TEUs in 2024 (PSA Singapore, 2025; "Tuas Mega Port Hits Milestone," 2025), underscoring how automation increases the port's operational efficiency and provides cost advantage and time advantage.

4. Policy Proposal and Evaluation

4.1 Policy proposal

4.1.1 Proposal 1: Subsidies to secondary ports

One of the most fundamental approaches to prevent PSA's monopoly power is to promote competition. In the face of fierce competition, the high concentration of throughput and hub lock-in effects in the Southeast Asian container transshipment market can be mitigated. Specifically, governments and regional authorities could provide targeted subsidies to secondary ports in Malaysia, Vietnam, and Indonesia that are actively investing in modern port infrastructure to support the transition from traditional to automated ports. By reducing the fixed costs of entry and expansion, such subsidies can eliminate barriers to entry, thereby increasing the number of intelligent ports entering the market. This can increase competition, reducing PSA's market power and preventing the formation of a monopoly.

4.1.2 Proposal 2: Transparency-Based Regulation

Even with greater regional competition, dominant ports may still retain significant market power. To address this, the Maritime and Port Authority of Singapore should require mandatory pricing transparency and cost disclosure for essential port services. In this case, transparency in PSA's cost structure reduces information asymmetry by allowing port users to identify whether price increases reflect legitimate AI-related cost recovery or increased profit margins driven by market power. Increased accountability raises the reputational and regulatory costs of unjustified price increases, thereby reducing PSA's pricing power. Hence, price will be close to marginal cost, reducing allocative inefficiency in the market.

4.1.3 Proposal 3: Cost-reflective price ceiling

Meanwhile, the Maritime and Port Authority of Singapore can impose a Cost-reflective price ceiling, setting a price ceiling explicitly related to the port's cost structure. To be more specific, the regulated price may cover efficient operating costs, capital expenditures related to automation investments, and depreciation based on asset lifespans. In addition, the port operator is permitted to earn a reasonable rate of return on invested capital. Therefore, this policy reduces the difference between price and marginal cost, reducing allocative inefficiency in the market.

4.2 Evaluation of the policies

4.2.1 Evaluation of the first proposal

The advantage of the first policy is that it can directly prevent the formation of a monopoly rather than merely monitor or regulate it by increasing competition in the Southeast Asian container transshipment market. The practicality is evident in Northern Europe. In this region, long-term public investment in automation has sustained strong inter-port competition, preventing the emergence of a single dominant monopoly port (*Bipartisan Infrastructure Law: Maritime Administration | MARAD*, 2021). Moreover, this policy will not distort price signals, making the market more natural.

However, a disadvantage is the substantial financial burden on the government of investing in automation. For example, in the United States, the 2021 Bipartisan Infrastructure Law allocated $17 billion for port infrastructure, with approximately $2.6 billion specifically designated for upgrading equipment and reducing air pollution, including automation technologies, imposing a substantial financial burden on the government.

4.2.2 Evaluation of the second proposal

Considering the second policy, transparency-based regulation is widely applied in other network industries such as electricity, water, airports, and telecommunications. For example, airport operators in many OECD countries are required to justify fee increases through cost disclosures, thereby constraining pricing power and increasing scrutiny from users, regulators, and the public, demonstrating the policy's feasibility *(COMPETITION COMMITTEE Series Roundtables on Competition Policy N° 35 2001 Price Transparency, n.d.)*. More importantly, it does not cap prices or restrict investment, preserving incentives for automation and innovation.

However, the disadvantage is its relatively low effectiveness, as it does not eliminate monopoly power on its own. If alternative ports remain weak, with low operating efficiency and stability, consumers will still choose PSA, even though it raises prices to achieve a higher profit margin. Moreover, information disclosure entails administrative costs and demands robust regulatory capacity.

4.2.3 Evaluation of the third proposal

Finally, Price-celling regulation is well established in capital-intensive infrastructure sectors such as utilities, railways, airports, and pipelines. For instance, in the UK, Ofgem's RPI-X price-cap framework, which allows cost recovery through depreciation and an allowed WACC-based return, has successfully lowered prices while preserving service quality and more than £35bn in network investment (Utility Regulator Northern Ireland, 2012). For PSAs, which are also capital-intensive infrastructure, the price-cap regulation should be effective, as in the case of Ofgem's. Meanwhile, carefully designed cost-reflective mechanisms can also preserve dynamic efficiency by protecting investment incentives.

The drawback is that it requires detailed cost assessments and strong regulatory expertise, as an inappropriate price cap can engender severe market failure. Therefore, price caps should be designed with a large time cost to ensure the effectiveness and appropriateness of this intervention.

5. Limitation & Conclusion

In all, multi-faceted microeconomic policy approach is essential as transition happens towards automation of ports, which represents a mixed factor for the growth of global economy. One limitation of this research is the reliance on theoretical framework with the lack of primary data, which can provide key insights to what decisions and changes are actually caused by automation in ports instead of other factors. Shown from the evolution of Singapore's Tuas Port, automation provides efficiency, reliability, and sustainability. However, these technological leaps pose significant structural risks: the potential for "Hub Lock-in" effects that grant significant market power and promotes regional monopolies. Microeconomic interventions—such as subsidizing competition in secondary ports and implementing cost-reflective price caps—are important to prevent harmful monopolistic behaviors and maintain allocative efficiency. Ultimately, the goal of port automation should not only be a tool of exploitation for monopolies, but the creation of a sophisticated and resilient supply chain for society.

6. AI Use Statement

AI were used in grammatical corrections. All ideas, analytical reasoning and direct text are from original work.

Reference

1. Bipartisan Infrastructure Law: Maritime Administration | MARAD. (2021). U.S. Department of Transportation. https://www.maritime.dot.gov/about-us/bipartisan-infrastructure-law-maritime-administration

2. COMPETITION COMMITTEE. (n.d.). *Series roundtables on competition policy N° 35 2001 price transparency*. OECD Publishing. https://www.oecd.org/content/dam/oecd/en/publications/reports/2001/09/price-transparency_546f6104/1cd2bc99-en.pdf

3. Environmental Benefits of Port Automation: Reducing Emissions and Energy Consumption | VS&B Containers. (2025). VS&B Containers. https://www.vsnb.com/environmental-benefits-port-automation-reducing-emissions-and-energy-consumption

4. Hand, M. (2026, January 29). World's largest automated terminal: PSA Tuas Port pioneering automation transformation with event-driven architecture. *Seatrade Maritime*. https://www.seatrade-maritime.com/terminals/world-s-largest-automated-terminal-psa-tuas-port-pioneering-automation-transformation-with-event-driven-architecture

5. International Transport Forum. (2021). *Container port automation: Impacts and*

6. Kottara, C. (2025). The impact of Artificial Intelligence in the shipping sector and the role of Total Quality Management. *Journal of Research in Business and Management*, 13(7), 42–48. https://doi.org/10.35629/3002-13074248

7. Knatz, G., Notteboom, T., & Pallis, A. (2022). *Container terminal automation: A global analysis and survey on decision-making drivers, benefits realized, and stakeholder support* (Final report, Project No. MF-2.2e). METRANS Transportation Consortium, University of Southern California.

8. Li, K., Gharehgozli, A., & Lee, J.-Y. (2023). Smart technologies and port operations: Optimal adoption strategy with network externality consideration. *Computers & Industrial Engineering*, 109557. https://doi.org/10.1016/j.cie.2023.109557

9. Miller, S. (2025, April 1). The impact of the Red Sea Crisis on global supply chains – Building resilience amid maritime security threats. *Sigma7*. https://www.s7risk.com/the-impact-of-the-red-sea-crisis-on-global-supply-chains/

10. Monopoly Power, Market Definition, and the Cellophane Fallacy. (2015, June 25). U.S. Department of Justice. https://www.justice.gov/archives/atr/monopoly-power-market-definition-and-cellophane-fallacy

11. MPA. (2024). Global connectivity. *Maritime and Port Authority of Singapore*. https://www.mpa.gov.sg/maritime-singapore/what-maritime-singapore-offers/global-hub-

port

12. MPA. (n.d.). *About MPA*. https://www.mpa.gov.sg/docs/mpalibraries/mpa-documents-files/comms-and-community/annual-report/mpa-ar24-full-book_fa.pdf?sfvrsn=2f89c7a0_1

13. MPA Integrated Annual Report 2022. (2022). *Maritime and Port Authority of Singapore*. https://www.mpa.gov.sg/staticfile/Cwp/assets/AR/2022/index.html

14. Notteboom, T., & Yim Yap, W. (n.d.). Port competition and competitiveness. VLIZ. https://www.vliz.be/imisdocs/publications/258122.pdf

15. PSA-admin. (2025, February 24). PSA Singapore celebrates 10 million TEU milestone at Tuas Port. *PSA Singapore*. https://www.singaporepsa.com/2025/02/24/psa-singapore-celebrates-10-million-teu-milestone-at-tuas-port

16. PSA Singapore. (n.d.). *Sustainability at PSA Singapore 2021*. https://www.singaporepsa.com/wp-content/uploads/2022/10/Sustainability-at-PSA-Singapore-2021.pdf

17. Red Sea Crisis Update: Route Alternatives & Cost Impacts. (2026, January 5). *DocShipper*. https://docshipper.com/shipping/red-sea-crisis-update-route-alternatives-cost-impacts/

18. Revision to PSA Singapore Container Terminals Price List. (2022, June 2). *X-Press Feeders*. https://www.x-pressfeeders.com/news/revision-to-psa-singapore-container-terminals-price-list

19. Schmidt, E. (2022, November 3). Why does the U.S. lag other nations so badly in the automation of its ports? *APM Research Lab*. https://www.apmresearchlab.org/10x-port-automation

20. Tuas mega port hits milestone of handling 10 million containers since opening. (2025). *ASMI*. https://www.asmi.com/en/news-updates/cm7k1zpbz0001lfx6gd6tsoal

21. Tsagkaris, P., & Moschovou, T. P. (2025). The impact of automation on the efficiency of port container terminals. *Future Transportation*, 5(4), 155. https://doi.org/10.3390/futuretransp5040155

22. United Nations Conference on Trade and Development. (2025, April 23). Shipping data: UNCTAD releases new seaborne trade statistics. *UNCTAD*. https://unctad.org/news/shipping-data-unctad-releases-new-seaborne-trade-statistics

23. United Nations Conference on Trade and Development. (2021). *Review of maritime transport 2021*. UNCTAD. https://unctad.org/publication/review-maritime-transport-2021

24. Utility Regulator Northern Ireland. (2012). Transmission and distribution RP5 price control: Response to the draft determination (Response from Northern Ireland Electricity Limited). *Utility Regulator*

(NI). https://www.uregni.gov.uk/files/uregni/mediafiles/RP5_Draft_Determination_rsponse_from_NIE_Ltd_0.pdf

25. Williams, M. (2025, March 4). PSA is using the latest automation technology to increase container throughput at the port of Tuas. *Automotive Logistics*. https://www.automotivelogistics.media/digitalisation/psa-is-using-the-latest-automation-technology-to-increase-container-throughput-at-the-port-of-tuas/216091

26. World Bank Group. (2024). *The Container Port Performance Index 2023: A comparable assessment of performance based on vessel time in port*. World Bank Open Knowledge Repository. https://openknowledge.worldbank.org/server/api/core/bitstreams/6cebb847-6f46-44e7-9533-12ac893b3693/content